

Tenzorske metode za Recommender Systems

Jakov Tušek, Nikola Mamić, Petar Matejčić

- Problem: korisniku preporučiti oznake za određeni artikl

$$Y : U \times I \times T \rightarrow \mathbb{R}$$

- $S \subseteq U \times I \times T$

$$D_S := \{(u, i, t_A, t_B) : (u, i, t_A) \in S \wedge (u, i, t_B) \notin S\}$$

$$p(\Theta | >_{u,i}) \propto p(>_{u,i} | \Theta) p(\Theta)$$

$$p(\Theta | \succ_{u,i}) \propto p(\succ_{u,i} | \Theta) p(\Theta)$$

$$p(\succ_{u,i} | \Theta) = \prod_{(u,i,t_A,t_B) \in D_S} p(t_A \succ_{u,i} t_B | \Theta)$$

$$p(\Theta | >_{u,i}) \propto p(>_{u,i} | \Theta) p(\Theta)$$

$$p(>_{u,i} | \Theta) = \prod_{(u,i,t_A,t_B) \in D_S} p(t_A >_{u,i} t_B | \Theta)$$

$$p(t_A >_{u,i} t_B | \Theta) = \sigma(y_{(u,i,t_A,t_B)}(\Theta))$$

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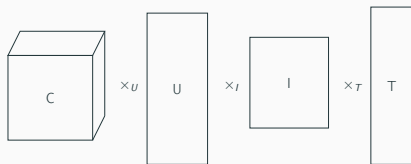
$$\Rightarrow \ln(p(\Theta | >_{u,i})) = \sum_{(u,i,t_A,t_B) \in D_S} \ln \sigma(\hat{y}_{u,i,t_A,t_B}(\Theta)) - \lambda_{\Theta} \|\Theta\|_F^2$$

$$\ln(p(\Theta | \mathcal{D}_{u,i})) = \sum_{(u,i,t_A,t_B) \in D_S} \ln \sigma(\hat{y}_{u,i,t_A,t_B}(\Theta)) - \lambda_{\Theta} \|\Theta\|_F^2$$

- Problem: D_S brzo raste
- Uniformno biramo $(u, i, t_A, t_B) \in D_S$

$$\nabla \ln(p(\Theta | \mathcal{D}_{u,i})) \approx \nabla \left[\ln \sigma(\hat{y}_{u,i,t_A,t_B}(\Theta)) - \lambda \|\Theta\|_f^2 \right]$$

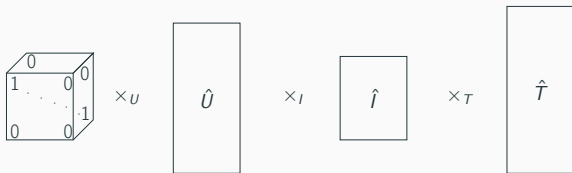
Tuckerova faktorizacija



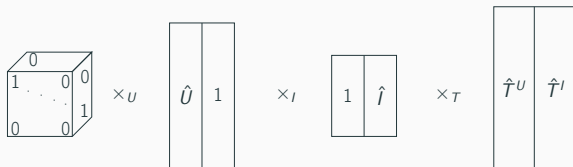
$$\hat{y}_{u,i,t}^{TD} = \sum_{u'} \sum_{i'} \sum_{t'} \hat{c}_{u',i',t'} \hat{u}_{u,u'} \hat{i}_{i,i'} \hat{t}_{t,t'}$$

$$\begin{aligned} \hat{C} &\in \mathbb{R}^{k_u \times k_i \times k_t}, & \hat{U} &\in \mathbb{R}^{|U| \times k_u} \\ \hat{I} &\in \mathbb{R}^{|I| \times k_i}, & \hat{T} &\in \mathbb{R}^{|T| \times k_t} \end{aligned}$$

CP dekompozicija



$$\hat{y}_{u,i,t}^{CP} = \sum_f^k \hat{u}_{u,f} \hat{i}_{i,f} \hat{t}_{t,f},$$

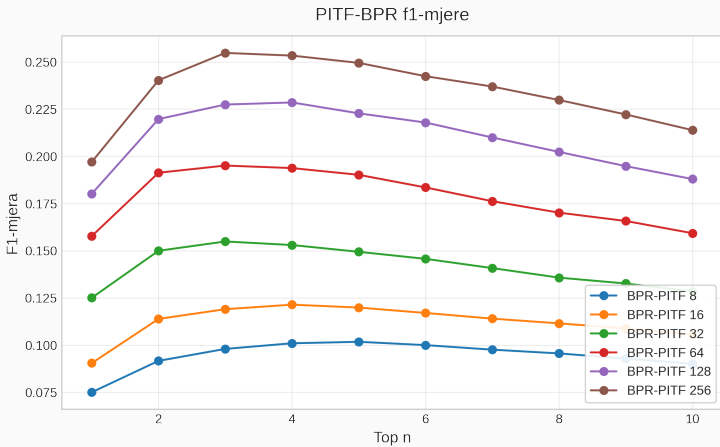


$$\hat{y}_{u,i,t} = \sum_f \hat{u}_{u,f} \hat{t}_{t,f}^U + \sum_f \hat{i}_{i,f} \hat{t}_{t,f}^I + \sum_f \hat{u}_{u,f} \hat{i}_{i,f}$$

$$\begin{aligned} \hat{U} &\in \mathbb{R}^{|U| \times k}, & \hat{I} &\in \mathbb{R}^{|I| \times k} \\ \hat{T}^U &\in \mathbb{R}^{|T| \times k}, & \hat{T}^I &\in \mathbb{R}^{|T| \times k} \end{aligned}$$

- 7-core dataset
- 4,304 korisnika, 7,475 filmova, 19,745 oznaka, 4,012,200 veza
- korisnički generirane oznake

Utjecaj parametra k - MovieLens



Utjecaj parametra λ - MovieLens

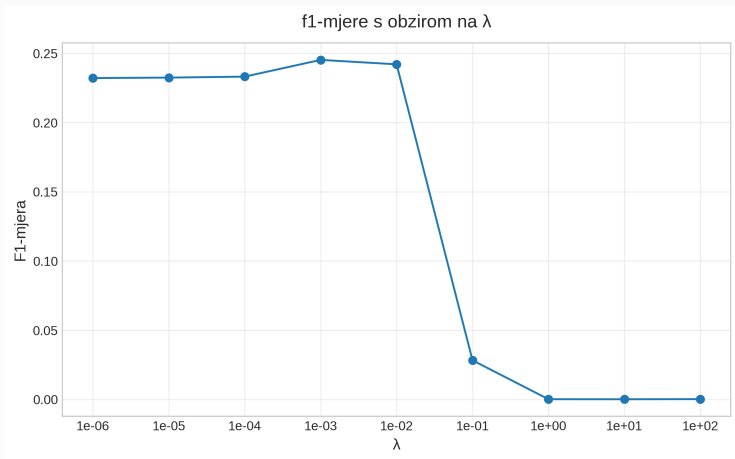
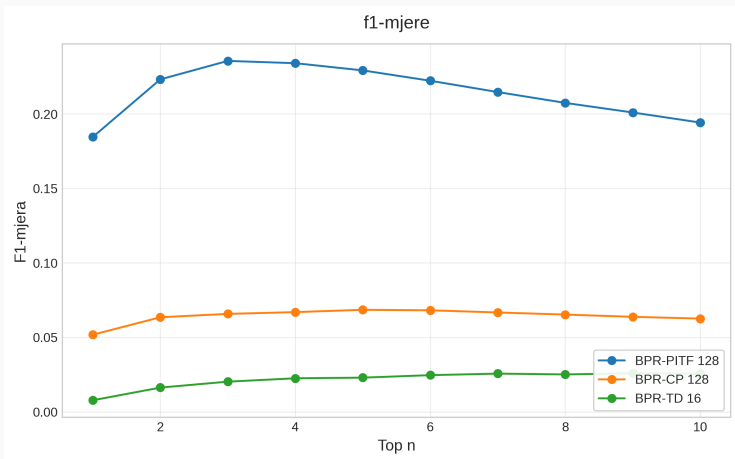


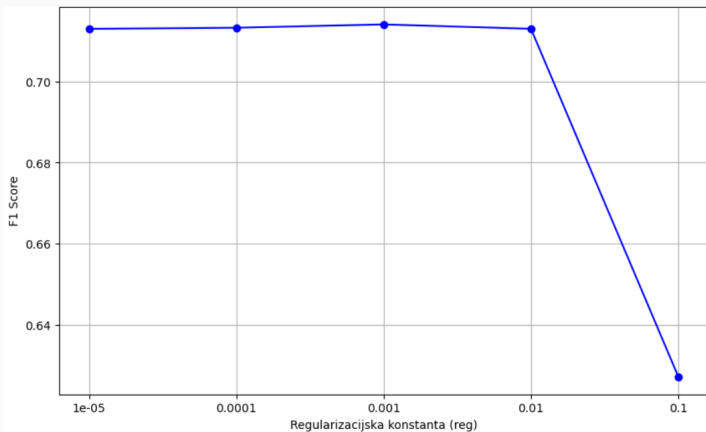
Figure 1: Caption

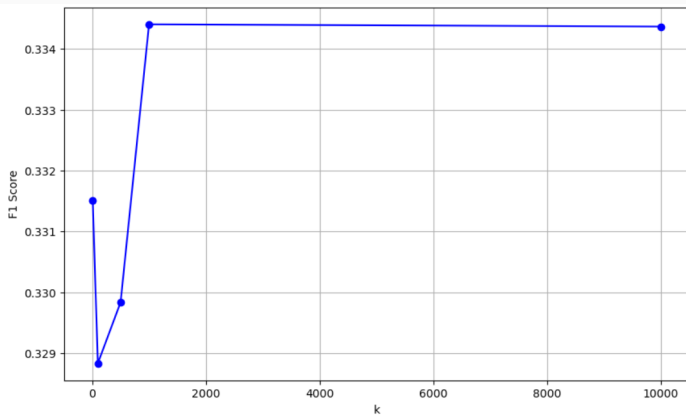
Usporedba modela - MovieLens



- 10-core dataset
- 17,000,000 ocjena
- 3 oznake (1-3, 4-6, 7-10)

PITF - libimseti



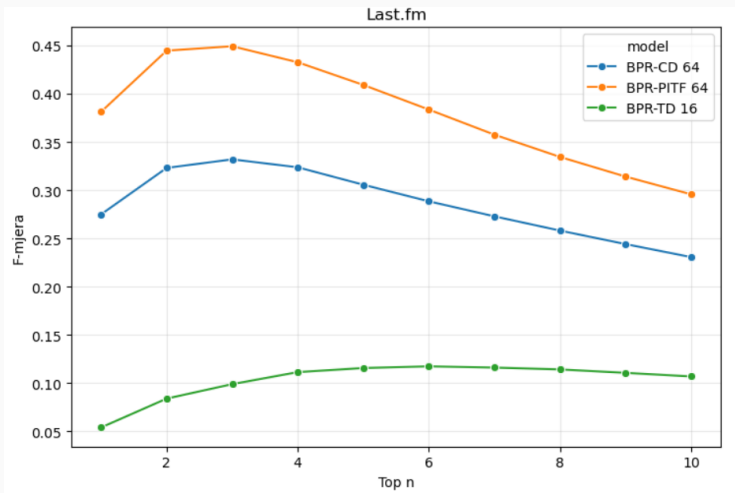


- 3-core dataset
- 1,458 korisnika, 5,074 web-dokumenata, 4,233 oznaka, 84,000 veza

Algoritam	F1@3
PITF	0.22
CP	0.04

- 7-core dataset
- 811 osoba, 2,386 glazbenika/grupa, 1,288 oznaka 111,149 veza

Usporedba modela - last.fm



Utjecaj parametra k - last.fm

